

Topology-Correcting Differentiable Triangle-Soup Reconstruction via Persistent Homology

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Abstract

Resampling priors can steer where a differentiable triangle-soup reconstruction spends its triangle budget, but our controlled study of that channel (supplementary material) found their topological value carried largely by spatial width, and no prior shape repairs one-dimensional features. We move topology from the allocation channel into the objective: $L = \text{photometric} + \lambda(t) L_{\text{topo}}$, where L_{topo} is an optimally-matched distance between the live surface’s persistence diagram and a fixed target diagram. Gradients flow through a pair-frozen backward: matched birth/death simplices re-expressed as closed-form circumradii of live sample positions (exact for Gabriel simplices — all significant pairs in practice), plus a recruitment term restoring the gradient optimal matching provably lacks when a target feature is missing; one ratio knob ρ sets the scale by gradient-norm calibration, and a curriculum proved unnecessary. On budget-starved synthetic scenes the loss is topology-specific where a norm-matched non-topological control is not: enclosed voids improve 4.0–7.9 $\times$ and — the channel’s distinguishing result — torus loops improve 2.3 $\times$ with zero phantom handles across seeds, while the control collapses a loop. The channels compose: loss plus the best prior reaches 7.6 $\times$ below baseline at the best Chamfer. Component counting (H0) is again matched by the generic control — this class needs no topological information. All verdicts replicate, with no per-shape tuning, on four external genus-known meshes — one a raw artist mesh ingested and certified rather than constructed — at 2.1–10.4 $\times$ and equal-or-better Chamfer, and the loss degrades gracefully (never below baseline) under sensor noise up to one sample spacing. Prescription: correct topology in the loss, allocate with a wide prior, and combine.

1. Introduction

Differentiable rasterization reconstructs a triangle soup from images by following photometric gradients [46], periodically resampling triangles under a fixed budget. Photometric losses are topologically blind: reconstructions that differ in genus, connectivity, or enclosed-void structure can be indistinguishable to Chamfer and Hausdorff while the bottleneck distance between persistence diagrams separates them by an order of magnitude (suppl. Table S1).

A decade of connectivity-free reconstruction — point splats, Gaussians, triangle soups — has treated topology errors as symptoms of geometric underprovision: represent the surface better, or allocate the primitive budget more wisely, and topology should follow. The allocation half is directly testable, and we tested it first — biasing where the resampler spends triangles in a controlled prior study (suppl. §A). It fails: a topological prior helps enclosed voids, but any equally-wide non-topological prior reproduces most of the gain, and no prior shape repairs loops — concentrating manufactures phantom handles, spreading never beats a random control.

The null result relocates the root cause: not where triangles are spent, but what the objective measures — a photometric loss is a local, pixel-space signal; a loop or an enclosed void is a global, categorical property of the surface. This paper therefore moves topology into the objective: a differentiable persistence term in the training loss, so gradients act on the birth/death values of the reconstruction’s topological features directly, rather than on where triangles respawn.

Three research hypotheses structure the study, each tested by a condition designed in advance. RH1 (diagnosis): topology errors originate in the training objective, not in primitive allocation — tested by the allocation study above. RH2 (specificity): a persistence term improves topology in ways no norm-matched non-topological regularizer through the identical channel

can reproduce — the control C2 (§3.5). RH3 (complementarity): objective and allocation are complementary channels, not substitutes — the composition C5. Concretely:

- A matched persistence loss with a pair-frozen backward (§3): exact persistence forward (GUDHI alpha complex); backward, each paired birth/death simplex re-expressed as the closed-form circumradius of live surface samples — Gabriel-exact for 100% of significant pairs on our shapes, so the frozen gradient is the true gradient almost everywhere.
- A recruitment term for missing features: optimal partial matching yields exactly zero gradient when a target feature has no live counterpart — the most important failure mode; recruitment attaches the nearest live bar instead. A controlled toy probes its known location-blindness; removing it in-loop collapses the loop-class win to baseline (§5).
- One calibrated knob: λ is set by a gradient-norm ratio ρ against the photometric loss; the response is flat over a $10\times$ range and a curriculum adds nothing — an ablation, not an assumption.
- Channel-controlled evaluation: every claim is tested against a norm-matched non-topological control acting through the identical channel (a repulsion on the same surface samples) — the allocation study’s control discipline, inherited.

The loss is topology-specific for enclosed voids ($4.0\text{--}7.9\times$ at Chamfer parity) and, unlike every prior we tested, for loops ($2.3\times$, zero phantom handles) — RH2 confirmed on the class allocation never won. The channels compose ($7.6\times$; RH3); components (H0) are again matched by the generic control — three channels agree the class carries no exploitable signal, sharpening RH1: topological guidance earns its keep where geometry and topology decouple.

2. Related work

A decade of gradient-based reconstruction: four positions on connectivity. OpenDR [32], neural mesh rendering [26], SoftRas [31], DIB-R [8] and nvdiffrast [28] established gradient flow from pixels to geometry; reconstruction work since splits on what that geometry should be. (i) Implicit fields — learned occupancy and signed distance [34, 39], volume-rendered from NeRF [35] through NeuS [49] to Neuralangelo [30] — are watertight and topologically consistent by construction (the strongest argument against explicit soups), paid for with dense field evaluation and post-hoc extraction. (ii) Fixed-connectivity templates deformed per scene [19, 48] keep manifoldness but freeze topology at the template’s genus. (iii) Differentiable iso-surfacing re-extracts the mesh

from a volumetric scaffold at every step [36, 43, 44] (Shape-as-Points bridges point soups to watertight implicits [40]) — topological freedom at grid cost. (iv) Connectivity-free primitive soups — point splatting [50], patch atlases [17], Gaussians [27] and their surface-accurate variants [18, 23], triangle splatting [6, 20], and DiffSoup [46] — are the fastest and the easiest to budget, but even their surface-minded members regularize geometry (alignment, flatness, connectivity forces), never a measured topology. (i) and (iii) solve topology representationally; (iv) had not addressed it. Cut by optimization philosophy rather than representation, the decade collapses to two occupied families — geometry-oriented objectives (photometric plus Eikonal/alignment/flatness terms: every method above) and sampling-oriented steering of where a fixed objective spends primitives (below) — and an empty third: topology-oriented objectives that measure the property and differentiate through it (suppl. Table S6 tabulates the decade under this cut). This paper fills the empty family at the soup’s budget and speed. We build on DiffSoup, whose in-loop prune/respawn resampler is exactly the mechanism our allocation-channel study (suppl. §A) steered with importance priors; here the resampler stays untouched and the objective changes. Vertex-domain optimizers such as large-steps preconditioning [37] are complementary — they change how vertices move, we change why.

The allocation channel: why priors alone fail. Three findings of our allocation-channel study set this paper’s problem. (1) Geometric metrics are topology-blind: on bisection-matched probes with Chamfer equal by construction, the discriminating-dimension bottleneck separates correct from broken topology by $35\text{--}40\times$ (suppl. Table S1). (2) For voids, a spread topological prior cuts the bottleneck $33\text{--}53\%$ below baseline — but a width-matched random control recovers most of that gain, leaving a small, shape-dependent residual: the prior’s value is carried largely by its width. (3) No prior shape repairs loops: the best torus arm is the non-topological wide control (.0267 vs the spread prior’s .0316), and the concentrated field manufactures phantom handles (4.4 significant loops vs the true 2). The composition condition C5 reuses that study’s strongest field (B4; suppl. §A).

Persistence in learning objectives. Persistent homology [13] with its stability guarantees [10] has been used as a trainable signal via differentiable sorting of filtration values [5], for topology-preserving image segmentation [9, 22], point-cloud and shape optimization [4, 41], 3D volume prediction [47], and

diagram-metric optimization with convergence analysis [7]; barcode-valued maps have since been given a general differential framework [29], and the per-step cost of persistence-guided descent attacked directly [38]. Nearest our setting, concurrent work brings persistence to reconstruction at other entry points: an image-space barcode term for Gaussian splatting [45], homology-guided camera placement [15] or a genus-matched template [14] in mesh inverse rendering, and a connected-component constraint on implicits fitted to point clouds [24]; none measures the evolving surface’s diagram inside image-based training, none under a primitive budget. Our backward pass is in this family — unlike work that back-propagates through filtration-value sorting or barcode coordinates in general position, we re-express each paired simplex’s birth/death as a closed-form circumradius of live surface samples — with three pipeline-specific ingredients. (i) Alpha-complex pairing over $\alpha \times \text{area}$ surface samples of a live triangle soup — not image grids or raw clouds — couples the diagram to the geometry actually being trained. (ii) A Gabriel-exactness gate: the circumradius re-expression equals the filtration value only for Gabriel simplices, so we verify it per pairing (relative 10^{-6}) — orthogonal to the standard pair-frozen device: it certifies the frozen gradient rather than assuming it. (iii) Recruitment for missing features: optimal partial matching has zero gradient exactly when a target feature is entirely absent — the worst failure mode — so we trade metric fidelity for a restored descent direction (§3), quantifying both its pathology and its necessity (§5).

Topology-aware geometry processing. Topology-aware reconstruction and repair predate differentiable pipelines [2, 25, 42], and budgeted remeshing has a long lineage [1, 3, 11, 16, 21]; our contribution is orthogonal — the budget mechanics stay DiffSoup’s, only the training signal becomes topology-aware.

3. Method: a matched persistence loss for triangle soups

3.1. Objective

DiffSoup trains with a photometric objective L_{photo} (opacity auxiliary + $0.8 L_1 + 0.2 \text{SSIM}$) [46], which we adopt unchanged: every ablation below isolates the topology channel, never photometric tuning. We add

$$L = L_{\text{photo}} + \lambda(t) L_{\text{topo}}(V), \quad (1)$$

where V are the soup’s vertex positions and L_{topo} compares the persistence diagram of the live surface to a fixed target bundle $\mathcal{T}$: the significant finite bars of

the ground-truth shape’s diagram, computed once at the sampling density M the loss will see (alpha values shift with spacing: density matching is a correctness requirement), normalized by the target’s bounding-box diagonal, and locked for all live evaluations.

3.2. Live diagram and pair-frozen backward

At each refresh we draw M surface samples ($M=2048$ unless the measurement-floor rule of §4 dictates a denser bundle) $X_j = \sum_i w_{ij} V[F(f_j), i]$ with faces f_j drawn $\sigma(\alpha) \times \text{area}$ -weighted and barycentric weights w_{ij} frozen; X is recomputed from live V every step, so gradients reach vertices while the combinatorics stay fixed. The alpha complex of X [12], computed exactly with GUDHI [33], yields persistence pairs (birth simplex, death simplex) per dimension. For a Gabriel simplex the alpha filtration value is its squared circumradius, so each paired bar’s coordinates are closed-form functions of a few sample positions, all exactly differentiable:

$$R_{\text{edge}} = \frac{1}{2} \|p_1 - p_0\|, \quad R_{\text{tri}} = \frac{\|u\| \|v\| \|u - v\|}{2 \|u \times v\|}, \quad (2)$$

$$R_{\text{tet}} = \|y\| \quad \text{with} \quad Ay = \frac{1}{2} \text{diag}(AA^T),$$

where $u = p_1 - p_0$, $v = p_2 - p_0$, and A stacks the rows $p_i - p_0$ (the circumcenter is $p_0 + y$). Edges carry H0 deaths and H1 births, triangles H1 deaths and H2 births, tetrahedra H2 deaths. We verify, per pairing, that the re-expressed value matches the filtration value (relative 10^{-6}); a pair failing this Gabriel test is detached — it keeps its (correct) value in the loss but propagates no gradient through that simplex, so an exactness failure can never inject a wrong gradient. In practice the gate never fires: 100% of significant pairs are Gabriel-exact on all six shapes’ clean clouds, with zero failures in the stress test’s 2,400 noisy refreshes (§5); the per-refresh count is logged, and stronger fallbacks (dropping pairs, subgradients, locally raising M) stay compatible. The pairing is treated as locally constant — the standard pair-frozen device [5, 7] — refreshed every $K=10$ steps and, necessarily, after every resampling event (which replaces the vertex and face tensors wholesale).

3.3. Matching, diagonal pressure, and recruitment

Let D be the live significant bars and $\mathcal{T}$ the target bars of one dimension. An optimal partial matching (Hungarian on the standard augmented cost; equivalently W_2) splits bars into matched pairs, unmatched

live, and unmatched target:

$$L_{\text{topo}} = \sum_{(i,j) \text{ matched}} [(b_i - b_j^*)^2 + (d_i - d_j^*)^2] + \sum_{i \text{ unmatched}} \left(\frac{d_i - b_i}{2}\right)^2 + L_{\text{recruit}}. \quad (3)$$

The first term pulls matched features to their target coordinates; the second pushes spurious features to the diagonal. The third exists because optimal matching has a blind spot: when a target feature is entirely missing, the cheapest transport diagonalizes both sides and the gradient is exactly zero. Recruitment restores a descent direction. Let $\mathcal{T}_{\text{miss}}$ be the unmatched target bars, processed in decreasing persistence order, and let the candidate pool $\mathcal{A}$ hold every live bar not claimed by the matching — unmatched significant bars and, crucially, sub-threshold bars, which matching never sees. Each missing target greedily claims its nearest pool bar, without replacement, under the same squared birth-death metric as the matched term:

$$L_{\text{recruit}} = \sum_{j \in \mathcal{T}_{\text{miss}}} [(b_{k(j)} - b_j^*)^2 + (d_{k(j)} - d_j^*)^2], \quad (4)$$

$$k(j) = \arg \min_{k \in \mathcal{A}_j} [(b_k - b_j^*)^2 + (d_k - d_j^*)^2],$$

where $\mathcal{A}_j$ is the pool remaining after earlier recruitments; recruited bars are excluded from the diagonal term and back-propagate through the same circumradius machinery. This deliberately departs from a metric-faithful W_2 ; §5 probes its known pathology (location-blindness) directly, and the C6 ablation shows the term is load-bearing in training.

3.4. Calibration instead of curriculum

$\lambda(t)$ ramps linearly from 20% to 50% of training, with peak set once by a gradient-norm ratio at the first active step: $\lambda_{\text{peak}} = \rho \|\nabla_V L_{\text{photo}}\| / \|\nabla_V L_{\text{topo}}\|$. The single knob ρ is dimensionless; $\rho=0.1$ throughout. The tail error is flat across $\rho \in \{0.03, 0.1, 0.3\}$, and an uncalibrated-schedule ablation (constant λ from step one) matches the ramped version — calibration, not scheduling, is what makes the knob safe. Alternative windows (10–40%, 30–60%; sphere, three seeds) land at $.0154 \pm .0017$ / $.0179 \pm .0019$ vs the default's $.0156 \pm .0013$ — within seed noise; 20:50 is conservative, not tuned.

3.5. Conditions

C0: baseline (photometric only). C1: $+L_{\text{topo}}$. C2: $+a$ short-range repulsion on the same frozen samples, calibrated with the same ρ — a norm-matched non-topological control through the identical channel, the

RH2 test (C2g: a gentler radius). C3: C1 without the ramp. C5: C1 + the best prior of our allocation study (the spread topological field B4; suppl. §A) — the RH3 test. C6: C1 with the recruitment term removed (pure matching + diagonal, all else identical) — does recruitment matter in-loop, or only for cold-start repair?

4. Experimental setup

Scenes and budgets. Synthetic COLMAP scenes with analytically known topology, rendered from 24–72 Fibonacci-sphere viewpoints at 200–256 px (every-8th test holdout $\Rightarrow$ 21–63 training views): sphere and cube (void class H2, $\beta=(1,0,1)$, budget $N=1200$), torus (loop class H1, $(1,2,1)$, $N=700$), two spheres (component class H0, $(2,0,2)$, $N=700$), and genus-2 double torus as a stretch case $((1,4,1)$, $N=2000$). Budgets are the allocation study's headroom points (suppl. §A) — tight enough that baseline topology is imperfect. Four external, genus-known meshes probe generality beyond the analytic family (§5.2). Three are watertight benchmarks: spot (genus 0) and bob (genus 1) from the Crane model repository (CC0), and the fan-disk CAD benchmark (genus 0) — rendered with the same pipeline (48 views, 200 px) at unchanged class budgets; a seed-0 C0 pilot confirms baseline imperfection (tails .043–.054, inside the analytic band). The fourth, the tom-yum pot (Fig. 2), probes the step every real deployment needs: a raw artist-authored mesh rather than a clean benchmark. As exported from Blender it is textbook triangle soup — 6,318 vertices in 9 overlapping open shells, mixed quad/12-gon faces, 50 boundary edges after exact welding, 232 non-manifold junction edges — with no trustworthy ground-truth topology (the same defect that excluded ShapeNet). An ingest-and-certify pass makes it usable: exact-duplicate weld, offset-solidification into a thin closed shell (exact unsigned distance field + marching cubes at iso ϵ), dropping 804 enclosed pocket shells, then certification by measurement (closed, edge-manifold, consistently oriented; exact simplicial homology; watertightness cross-check; bit-identical rebuilds), yielding one body of genus 3, $\beta=(1,6,1)$ exact (190,990 vertices), rendered and budgeted as a void-class shape ($N=1200$, 48 views, 200 px). Where the benchmark trio's topology is known by construction, the pot's is known by certificate — the pipeline real scans will need (challenge inventory: suppl. Table S4). The pot also exercises the measurement floor of §6 prospectively: no H1 loop of the certified shell (handle bores, flue) clears the $6r_{\text{med}}$ floor at any working M , while the narrow flue mouth caps at small α , making the flue interior an enclosed chamber — an H2 void on the pot axis, first clearing the floor at $M=8192$ (margin 1.20–1.23 $\times$

across five sampling seeds). The pre-registered floor rule builds its target bundle there — the study’s only protocol deviation — and the loss samples the live surface at the same density. The bundle carries exactly one significant bar (H2), stable to re-sampling ($\sim 10^{-5}$).

Protocol. 2,500 steps; resampling every 100 steps (untouched); $\rho=0.1$, ramp 0.2:0.5, refresh $K=10$, $M=2048$; decisive shapes (sphere, torus) run 5 seeds with the full condition set (C0/C1/C2/C3/C5, plus the recruitment ablation C6), replication shapes (cube, two spheres) 3 seeds (C0/C1/C2), double torus 2 seeds. A sensor-noise stress test (C7/C7h = C1 + noise; decisive shapes, 3 seeds) perturbs the cloud the loss plan sees with i.i.d. Gaussian noise at every refresh, $\sigma = 0.5\%/1\%$ of the target diagonal ($\approx 0.5/1.0$ median sample spacings at $M=2048$): the plan decides on the noisy observation, the pulls apply to the true live positions. The torus restricts the loss to H1 (its own void bar is sub-threshold at $M=2048$ — a loss must not act on features it cannot reliably measure); the double torus likewise targets its two measurable loops of four. The generality wave runs C0/C1/C2, three seeds; bob (fat tube) has both loops and void significant at $M=2048$ and runs the full bundle; the pot runs at the floor-rule density above.

Metrics and verdict. Primary: bottleneck distance to the target diagram in the shape’s discriminating dimension, tail-averaged over the last trajectory dumps, seed-averaged (20k evaluation samples, target-normalized). Secondary: significant-feature counts (phantom check) and Chamfer parity. Verdict rule: C1 must beat C0 and every norm-matched control at Chamfer parity — a topological win a dumb regularizer can match is not a topological win. Chamfer parity, formally: the passing arm’s seed-mean Chamfer must satisfy $\text{Ch}(C1) \leq 1.15 \text{Ch}(C0)$; the 15% headroom absorbs seed variance and CUDA nondeterminism (quantified in §7) and in practice never did any work — every passing arm’s Chamfer was at or below baseline.

5. Results

A gate first: the loss alone (no photometric term, Adam on point positions) repairs all four controlled probes — a punctured sphere’s void closes ($8.8\times$), a spurious handle dies at the diagonal, mis-spaced components separate ($1229\times$), and a bridged merge with no significant live H0 bar — the case where optimal matching provably has zero gradient — is severed ($450\times$, $\beta_0 1\rightarrow 2$); the predicted location-blind pathology did not materialize (suppl. §C, Fig. S5).

5.1. The C-matrix

Table 1 reports tail bottleneck (mean $\pm$ sd over seeds) in each shape’s discriminating dimension; Chamfer in the text where it matters; Fig. 1 shows the full training trajectories for the two decisive shapes.

Voids (H2): topology-specific, large. Sphere $4.0\times$ (16.2σ Welch vs C0), cube $7.9\times$ (18.5σ), both at equal-or-better Chamfer (sphere 0.46 vs 0.50; cube 0.92 vs 1.05). The norm-matched control is destructive, not merely weaker: it pins the void’s death at its repulsion equilibrium ($2.2\times$ worse than baseline, identical across seeds) and erases the cube’s void outright. Nor is this control-strength dependent (§7): the ordering is $C1 < C0 < C2g < C2$ — no tested strength reproduces any of the gain (per-seed spread: suppl. Fig. S3; pin-and-drag mechanism: §6).

Loops (H1): the channel’s distinguishing result. No prior shape won this class (suppl. §A). The loss cuts torus H1 error $2.3\times$ with zero phantom handles in all five seeds (count trajectories flat at the true $\beta_1=2$, suppl. Fig. S4), while the control collapses one loop by step ~ 800 and never recovers it. Metric pressure on birth/death values suits 1-D features in a way budget allocation does not (mechanism: §6); voids, whose support is the whole shell, respond to both channels — why the composition (C5) wins everywhere.

Components (H0): not topological — third channel, same answer. C1 reduces the (already small) baseline error $9.5\times$, but the generic control matches it on the diagram metric (.0008 vs .0007); the topological increment appears only in Chamfer (0.54 vs 1.94). Three channels now agree on the feature class: β_0 structure is carried by the count and spatial separation of shells — properties any spreading pressure improves and the photometric term already infers from images.

Composition (C5). Loss + spread prior is best everywhere tested — sphere .0082 ($7.6\times$ vs C0, $\approx 2\times$ vs C1 alone, $\approx 4.4\times$ below the prior alone), torus .0133 — with the best sphere Chamfer; on the torus its Chamfer (.502) trails only C3’s (.495). Allocation and value-tuning are complementary, not substitutes.

Stretch case (genus 2): saturated, honestly. On the double torus every arm returns a bottleneck of exactly .0264: the two tube loops are unexpressable at feasible budgets, so the case measures the representation’s ceiling, not the channel’s (full account: suppl. §B; floors quantified in §6).

Table 1. Tail bottleneck to target (lower is better); $\times$ = reduction vs C0. Verdict: C1 < C0 and C1 < every control at Chamfer parity.

shape (dim)	C0	C1 loss	C2 control	C3 no-ramp	C5 loss+prior
sphere (H2)	.0623 $\pm$.0063	.0156 $\pm$.0013 (4.0 $\times$)	.1372 (worse)	.0151 $\pm$.0004	.0082 $\pm$.0007 (7.6 $\times$)
cube (H2)	.0582 $\pm$.0046	.0074 $\pm$.0011 (7.9 $\times$)	.1346 (β_2 1 $\rightarrow$ 0)	—	—
torus (H1)	.0424 $\pm$.0031	.0180 $\pm$.0016 (2.3 $\times$)	.0457 (β_1 2 $\rightarrow$ 1)	.0155 $\pm$.0015	.0133 $\pm$.0004 (3.2 $\times$)
two spheres (H0)	.0065 $\pm$.0008	.0007 $\pm$.0002 (9.5 $\times$)	.0008 (ties C1)	—	—
double torus (H1)	.0264 $\pm$.0000	.0264 $\pm$.0000 (saturated)	.0264 $\pm$.0000	—	—

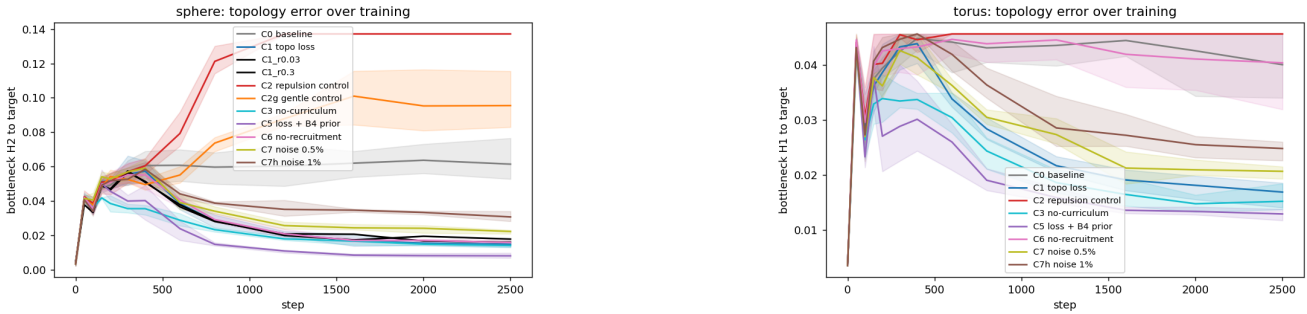


Figure 1. Topology error over training: bottleneck to target in the discriminating dimension (line = seed mean, band = seed min-max). Left, sphere (H2): every loss arm (C1, C3, C5, C6, both ρ variants, noise pair C7/C7h) converges to a fraction of baseline; the norm-matched repulsion control C2 is destructive, and its gentle variant C2g still ends worse than baseline. Right, torus (H1), the class no prior won: same ordering, C5 best; noise arms descend shallower but never approach baseline; the no-recruitment ablation C6 rides it.

Ablations. ρ -response flat over $\{0.03, 0.1, 0.3\}$ (.0181/.0172/.0177, matched seed-0 sweep, sphere; the five-seed $\rho=0.1$ arm is Table 1’s C1); C3 (no curriculum) $\approx$ C1 on both decisive shapes and off the analytic family (bob: C3 .0171 $\pm$.0026 vs C1 .0208 $\pm$.0008). Overhead, honestly: the loss adds a fixed ≈ 46 s per 2,500-step run (refresh 174ms at $M=2048$ every 10 steps) — more than the 33–41s photometric baseline on these deliberately tiny scenes, but the cost scales $O(M \log M)$ per refresh, not with scene or image complexity, so it amortizes where rendering dominates; no trainable parameters are added.

Recruitment is load-bearing for loops (C6). Removing the recruitment term (pure matching + diagonal, all else identical) collapses the torus win entirely: .0411 $\pm$.0038, statistically baseline (0.6 σ from C0, 12.4 σ worse than C1) at baseline Chamfer — though the loops still exist (#sig H1 = 2 in every seed; Fig. 1, right: the C6 curve rides the baseline). The logs locate the mechanism: one torus loop is significant at every refresh, the other chronically below threshold (all five seeds, all 200 refreshes) — recruitment is the sole gradient path to the second loop; on this class it is the mechanism, not a cold-start device. On the sphere the

void is always live and matched, recruitment never fires (0 of 200 refreshes, all seeds), so C6 is loss-identical to C1 by construction; its tail (.0166 vs .0156, 1.5 σ) measures pure run-to-run noise (§7).

Sensor-noise stress (C7/C7h). With the plan’s cloud perturbed at every refresh ($\sigma = 0.5\%/1\%$ of the diagonal $\approx 0.5/1.0$ median sample spacings, §4), the loss degrades gradually and never harms: torus .0210 $\pm$.0018 / .0259 $\pm$.0022 (2.0 $\times$ /1.6 $\times$ below baseline, vs C1’s 2.3 $\times$), sphere .0235 $\pm$.0009 / .0328 $\pm$.0013 (2.7 $\times$ /1.9 $\times$, vs 4.0 $\times$) — at equal-or-better Chamfer, with the correct significant-feature count in every run (Fig. 1) and zero Gabriel-gate failures in all 2,400 noisy refreshes (further internals: suppl. §B).

5.2. Generality: external genus-known meshes

Everything above is on one analytic family, so we repeat the verdict protocol — C0/C1/C2, three seeds, class budgets, ρ and ramp unchanged, density fixed by the floor rule (§4) — on four external genus-known meshes: spot (0, organic), bob (1, organic), fandisk (0, CAD), and the tom-yum pot (3, artist-authored, certified by ingest; §4). No per-shape tuning of any kind. All four pass the verdict rule at equal-or-better Chamfer (Table 2; trajectories in suppl. Fig. S2), and the con-



Figure 2. The tom-yum pot: an artist-authored Blender model (raw export: 9 open shells, 232 non-manifold junction edges) ingested and certified to a single genus-3 shell, $\beta=(1,6,1)$ exact (§4). Right: cutaway — the narrow flue mouth caps at small α , so the flue interior reads as an enclosed chamber: the on-axis H2 void the floor rule selects at $M=8192$ (all H1 loops sit below the floor).

trol’s failure modes replicate exactly: repulsion erases the void on all three H2 shapes ($\beta_2 1 \rightarrow 0$, all seeds; on the tom-yum pot at $1.36\times$ worse Chamfer) and collapses a loop on the genus-1 shape ($\beta_1 2 \rightarrow 1$, all seeds).

Four observations. The H1 claim survives the family change: bob halves loop error with zero phantom handles in every seed (26.1σ), replicating torus. Sharp-featured CAD benefits most: fandisk’s $10.4\times$ exceeds every analytic-family effect. Complex organic geometry exposes a count-level limit: spot’s baseline manufactures a phantom void in two of three seeds (final $\# \text{sig H2} = 2, 1, 2$); the loss clears it in two seeds but retains one spurious bar in the third ($1, 2, 1$) — value repair is reliable, count repair on sub-budget organic detail (legs, horns at $N=1200$) is an improvement, not a guarantee; spot’s C1 tail is pinned across seeds (sd 0 at .0237) — a geometric floor of the double-torus kind, far milder. And the pipeline reaches raw artist content: the tom-yum pot (Fig. 2) extends the wave in input class (a raw Blender export, certified rather than constructed, §4) and observable (an H2 void clearing the floor only at $M=8192$) at once — both chosen by the pre-registered rule: the dress rehearsal for scan data, whose ground truth must likewise be certified. The loss cuts the void’s value error $3.9\times$ at Chamfer parity ($0.99\times$ baseline); the 80.1σ Welch statistic reflects an unusually tight seed spread at $n=3$ (C1 sd .0002), so the honest summary is non-overlapping seed ranges, with the effect size inside the study’s $2.1\text{--}10.4\times$ span. The control destroys the void ($\beta_2 1 \rightarrow 0$ in all three seeds) at $1.36\times$ worse Chamfer; its bottleneck pins at the unmatched-target bound (the target bar’s half-life, .0223), so the count — not the meaningless value gap (0.8σ) — is the verdict. Two caveats transfer: the baseline already reads the correct count (a value-accuracy win, like bob), and ground truth is certified, not constructed.

6. Mechanism and discussion

A local signal cannot carry a global property — sparse tugs can. A pixel’s photometric gradient reaches only the few vertices rasterizing to it; topology is global and categorical — a wrong genus is not approached smoothly, and the blindness probes quantify the gap (Chamfer equal by construction, bottleneck apart $35\text{--}40\times$; suppl. Table S1) — so no geometric proxy stands in for the measurement. The persistence term supplies it with a usable gradient. Each significant pair back-propagates through the 3–7 sample positions of its birth and death simplices: precise but sparse pulls, iterated by re-pairing — the toys show surgery “walking” around a defect a few points at a time; the resampling prior is the opposite — broad, indirect, budget-limited. The results sort cleanly along this axis: 1-D loops, which allocation can only over- or under-tessellate, respond to value pressure (H1 win); wide 2-D shells benefit from both (prior helps, loss helps more, composition best); 0-D structure needs neither (H0 null, three channels).

Why the control damages topology. The norm-matched repulsion imposes a length-scale floor on the sampled surface: the void’s death pins at that equilibrium (identical across seeds), then drags, while near-diagonal phantom bars accumulate (diagram trajectories: suppl. Fig. S1), and thin structures collapse (β_2 on the cube, β_1 on the torus). The loss has no length-scale constraint — it pulls the death coordinate itself. The control argument sharpens: the same gradient budget through the same channel without topological information is not neutral but harmful — C1’s wins cannot be an artifact of extra regularization.

Measurability at the loss’s density. A persistence loss can only enforce features that clear the significance floor at its sampling density M : the torus void and the double torus’s two tube loops do not, at $M=2048$. We treat this as a principle, not a nuisance: the target bundle is built (and audited) at the same M , sub-floor features are excluded from the objective, and the double torus is evaluated against its two measurable loops — all the reconstruction itself can express at feasible budgets.

The floor is quantifiable: a bar is significant iff its lifetime exceeds $6r_{\text{med}}(M)$ (r_{med} = median nearest-neighbour spacing), and r_{med} follows the uniform-sampling law $\propto M^{-1/2}$ (measured .0090 $\rightarrow$.0028 over $M=2048 \rightarrow 20000$; ratio 3.16 vs $\sqrt{9.77}=3.13$). The double torus’s tube loops live at lifetime .045–.046 on the clean surface: at $M=2048$ that is $0.83\text{--}0.85\times$ the

Table 2. Generality wave: tail bottleneck to target (mean $\pm$ sd, 3 seeds); $\times$ = reduction vs C0; σ = Welch. Same verdict rule as Table 1.

shape (dim)	C0	C1 loss	C2 control	verdict
spot (H2)	.0521 $\pm$.0081	.0237 $\pm$.0000 (2.2 $\times$)	.0652 (β_2 1 $\rightarrow$ 0)	PASS (6.1 σ)
bob (H1)	.0426 $\pm$.0012	.0208 $\pm$.0008 (2.1 $\times$)	.0437 (β_1 2 $\rightarrow$ 1)	PASS (26.1 σ)
fandisk (H2)	.0538 $\pm$.0022	.0052 $\pm$.0008 (10.4 $\times$)	.0571 (β_2 1 $\rightarrow$ 0)	PASS (35.7 σ)
tom-yum pot (H2)	.0221 $\pm$.0003	.0057 $\pm$.0002 (3.9 $\times$)	.0223 (β_2 1 $\rightarrow$ 0)	PASS (80.1 σ)

floor — invisible by a 15% margin, clearing it only from $M \approx 3 \times 10^3$ — while at the evaluation density ($M=20000$, floor .0171) they clear it 3.1 $\times$; the binding constraint there is the representation (the soup never expresses the tubes), so raising M alone cannot close the gap. The torus quantifies the interaction with noise: its second loop clears the clean floor by only 1.33 $\times$, a margin the live cloud’s noise consumes — why it is chronically sub-threshold in training and recruitment is its only gradient path (C6). Recruitment is thus how the loss reaches below its own significance floor on the live side: landscape-guided in practice (suppl. §C), with the location-blindness caveat for targets beyond any achievable detour. Suppl. Table S3 tabulates counts across M (at 4096 all four double-torus loops clear); the tom-yum pot ran the rule prospectively (§4), and the in-loop result (Table 2) validated it. Adaptive density is the natural extension; the trade is explicit — persistence cost grows $O(M \log M)$.

Prescription. Correct topology in the loss; allocate with a wide prior; combine. Curriculum schedules are unnecessary once the loss scale is calibrated against the photometric gradient.

7. Limitations

Synthetic, closed, single-machine. All quantitative claims are on synthetic renders of closed surfaces with known topology, on one GPU: an analytic family plus four external genus-known meshes (§5.2; one a raw artist mesh, certified — the intended path for scans), short of a public benchmark. A real open-surface demonstration remains future work: scan boundaries contaminate H1 and no clean ground-truth diagram exists; of the concrete paths (boundary-aware persistence, filtering boundary-born bars, template-aligned initialization), bar-filtering is the pragmatic near-term step; a synthetic open-surface stress test comes first.

Density floor. The loss cannot see features below the significance floor at its sampling density (torus void; two of four double-torus loops at $M=2048$); the bundle audit makes this visible, but it is a real ceiling on the objective.

Fixed target diagram. The target bundle presumes known ground-truth topology (a prior scan or template; not blind capture). Template-free designs are untested here: total-persistence minimization, a target-genus penalty on the significant-H1 count (what we would try first), or statistical priors over a class’s diagrams.

Control strength. The repulsion control is aggressive (it destroys topology at calibrated magnitude); a gentler variant at half radius is still 1.6 $\times$ worse (sphere: .0972 vs .0623), so verdicts do not hinge on control tuning; control design still deserves more variety than one family at two strengths.

Nondeterminism. DiffSoup’s CUDA training is not bit-reproducible: divergence enters at resampling, where borderline prune/split decisions flip under floating-point drift. Tail CVs run 7–10% for baseline and loss arms alike; a loss-identical re-run pair (sphere C6, C1 by construction) landed 6% apart — pure noise. All statements are seed-averaged with Welch statistics.

Recruitment guarantees. Recruitment’s self-correction premise (unproductive edits saturate at the local detour scale) is validated empirically, not proven; pathological geometries could defeat it — a failure-case study is future work.

8. Conclusion

We moved topology out of the resampler’s allocation channel into the training objective, carried by a matched persistence loss with a per-pairing Gabriel-certified backward, a recruitment term restoring the gradient optimal matching provably lacks for missing features (load-bearing for loops), channel-controlled evaluation — every gain beats a norm-matched control at Chamfer parity — and a quantified account of the measurement ($6r_{\text{med}}(M)$) and representation (N) floors. The hypotheses hold — root cause in the objective (RH1), signal specifically topological (RH2), channels composing (RH3) — and the verdicts replicate on external genus-known meshes (one a raw artist mesh, certified), degrading gracefully under noise. Prescription: correct topology in the loss, allocate with a wide prior, combine. Next: open surfaces, real scans.

References

- [1] Pierre Alliez, David Cohen-Steiner, Olivier Devillers, Bruno Lévy, and Mathieu Desbrun. Anisotropic polygonal remeshing. *ACM Transactions on Graphics*, 22(3):485–493, 2003. 3
- [2] Marco Attene. A lightweight approach to repairing digitized polygon meshes. *The Visual Computer*, 26(11):1393–1406, 2010. 3
- [3] Mario Botsch and Leif Kobbelt. A remeshing approach to multiresolution modeling. In *Eurographics/ACM SIGGRAPH Symposium on Geometry Processing (SGP)*, 2004. 3
- [4] Rickard Brüel-Gabrielsson, Vignesh Ganapathi-Subramanian, Primoz Skraba, and Leonidas J. Guibas. Topology-aware surface reconstruction for point clouds. *Computer Graphics Forum*, 39(5):197–207, 2020. 2
- [5] Rickard Brüel-Gabrielsson, Bradley J. Nelson, Anjan Dwaraknath, and Primoz Skraba. A topology layer for machine learning. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 1553–1563, 2020. 2, 3
- [6] Nathaniel Burgdorfer and Philippos Mordohai. Radiant triangle soup with soft connectivity forces for 3D reconstruction and novel view synthesis. *arXiv:2505.23642*, 2025. 2
- [7] Mathieu Carrière, Frédéric Chazal, Marc Glisse, Yuichi Ike, Hariprasad Kannan, and Yuhei Umeda. Optimizing persistent homology based functions. In *International Conference on Machine Learning (ICML)*, 2021. 3
- [8] Wenzheng Chen, Jun Gao, Huan Ling, Edward J. Smith, Jaakko Lehtinen, Alec Jacobson, and Sanja Fidler. Learning to predict 3D objects with an interpolation-based differentiable renderer. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. 2
- [9] James R. Clough, Nicholas Byrne, Ilkay Oksuz, Veronika A. Zimmer, Julia A. Schnabel, and Andrew P. King. A topological loss function for deep-learning based image segmentation using persistent homology. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(12):8766–8778, 2022. 2
- [10] David Cohen-Steiner, Herbert Edelsbrunner, and John Harer. Stability of persistence diagrams. *Discrete & Computational Geometry*, 37(1):103–120, 2007. 2
- [11] Marion Donyach, David Vanderhaeghe, Loïc Barthe, and Mario Botsch. Adaptive remeshing for real-time mesh deformation. In *Eurographics 2013 — Short Papers*, 2013. 3
- [12] Herbert Edelsbrunner and Ernst P. Mücke. Three-dimensional alpha shapes. *ACM Transactions on Graphics*, 13(1):43–72, 1994. 3
- [13] Herbert Edelsbrunner, David Letscher, and Afra Zomorodian. Topological persistence and simplification. *Discrete & Computational Geometry*, 28(4):511–533, 2002. 2
- [14] Xiang Gao, Xinmu Wang, Xiaolong Wu, Jiazhi Li, Jingyu Shi, Yu Guo, Yuanpeng Liu, Xiyun Song, Heather Yu, Zongfang Lin, and Xianfeng David Gu. Inverse rendering for high-genus surface meshes from multi-view images. *arXiv:2511.18680*, 2025. 3DV 2026. 737, 738, 739, 740, 741, 742, 743, 744, 745, 746, 747, 748, 749, 750, 751, 752, 753, 754, 755, 756, 757, 758, 759, 760, 761, 762, 763, 764, 765, 766, 767, 768, 769, 770, 771, 772, 773, 774, 775, 776, 777, 778, 779, 780, 781, 782, 783, 784, 785, 786, 787, 788, 789, 790, 791, 792, 793
- [15] Xiang Gao, Xinmu Wang, Yuanpeng Liu, Yue Wang, Junqi Huang, Wei Chen, and Xianfeng Gu. Inverse rendering for high-genus 3D surface meshes from multi-view images with persistent homology priors. *arXiv:2601.12155*, 2026. ICASSP 2026. 3
- [16] Michael Garland and Paul S. Heckbert. Surface simplification using quadric error metrics. In *Proceedings of SIGGRAPH 97*, pages 209–216, 1997. 3
- [17] Thibault Groueix, Matthew Fisher, Vladimir G. Kim, Bryan C. Russell, and Mathieu Aubry. AtlasNet: A papier-mâché approach to learning 3D surface generation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018. 2
- [18] Antoine Guédon and Vincent Lepetit. SuGaR: Surface-aligned Gaussian splatting for efficient 3D mesh reconstruction and high-quality mesh rendering. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5354–5363, 2024. 2
- [19] Rana Hanocka, Gal Metzer, Raja Giryes, and Daniel Cohen-Or. Point2Mesh: A self-prior for deformable meshes. *ACM Transactions on Graphics*, 39(4), 2020. 2
- [20] Jan Held, Renaud Vandeghen, Adrien Delière, Abdullah Hamdi, Silvio Giancola, Anthony Cioppa, Andrea Vedaldi, Bernard Ghanem, Andrea Tagliasacchi, and Marc Van Droogenbroeck. Triangle splatting for real-time radiance field rendering. *arXiv:2505.19175*, 2025. 2
- [21] Hugues Hoppe. Progressive meshes. In *Proceedings of SIGGRAPH 96*, pages 99–108, 1996. 3
- [22] Xiaoling Hu, Fuxin Li, Dimitris Samaras, and Chao Chen. Topology-preserving deep image segmentation. In *Advances in Neural Information Processing Systems 32 (NeurIPS)*, 2019. 2
- [23] Binbin Huang, Zehao Yu, Anpei Chen, Andreas Geiger, and Shenghua Gao. 2D Gaussian splatting for geometrically accurate radiance fields. In *ACM SIGGRAPH 2024 Conference Papers*, 2024. 2
- [24] Anushrut Jignasu, Ethan Herron, Zhanhong Jiang, Soumik Sarkar, Chinmay Hegde, Baskar Ganapathysubramanian, Aditya Balu, and Adarsh Krishnamurthy. STITCH: Surface reconstruction using implicit neural representations with topology constraints and persistent homology. *arXiv:2412.18696*, 2024. 3
- [25] Tao Ju. Robust repair of polygonal models. *ACM Transactions on Graphics*, 23(3):888–895, 2004. 3
- [26] Hiroharu Kato, Yoshitaka Ushiku, and Tatsuya Harada. Neural 3D mesh renderer. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3907–3916, 2018. 2

- [27] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3D Gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42(4):139:1–139:14, 2023. [2](#) 851
- [28] Samuli Laine, Janne Hellsten, Tero Karras, Yeongho Seol, Jaakko Lehtinen, and Timo Aila. Modular primitives for high-performance differentiable rendering. *ACM Transactions on Graphics*, 39(6):194:1–194:14, 2020. [2](#) 852
- [29] Jacob Leygonie, Steve Oudot, and Ulrike Tillmann. A framework for differential calculus on persistence barcodes. *Foundations of Computational Mathematics*, 22:1069–1131, 2022. [3](#) 853
- [30] Zhaoshuo Li, Thomas Müller, Alex Evans, Russell H. Taylor, Mathias Unberath, Ming-Yu Liu, and Chen-Hsuan Lin. Neuralangelo: High-fidelity neural surface reconstruction. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. [2](#) 854
- [31] Shichen Liu, Tianye Li, Weikai Chen, and Hao Li. Soft rasterizer: A differentiable renderer for image-based 3D reasoning. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 7708–7717, 2019. [2](#) 855
- [32] Matthew M. Loper and Michael J. Black. OpenDR: An approximate differentiable renderer. In *European Conference on Computer Vision (ECCV)*, pages 154–169, 2014. [2](#) 856
- [33] Clément Maria, Jean-Daniel Boissonnat, Marc Glisse, and Mariette Yvinec. The Gudhi library: Simplicial complexes and persistent homology. In *International Congress on Mathematical Software (ICMS)*, pages 167–174, 2014. [3](#) 857
- [34] Lars Mescheder, Michael Oechsle, Michael Niemeyer, Sebastian Nowozin, and Andreas Geiger. Occupancy networks: Learning 3D reconstruction in function space. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019. [2](#) 858
- [35] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tanicik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. NeRF: Representing scenes as neural radiance fields for view synthesis. In *European Conference on Computer Vision (ECCV)*, 2020. [2](#) 859
- [36] Jacob Munkberg, Jon Hasselgren, Tianchang Shen, Jun Gao, Wenzheng Chen, Alex Evans, Thomas Müller, and Sanja Fidler. Extracting triangular 3D models, materials, and lighting from images. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8280–8290, 2022. [2](#) 860
- [37] Baptiste Nicolet, Alec Jacobson, and Wenzel Jakob. Large steps in inverse rendering of geometry. *ACM Transactions on Graphics*, 40(6), 2021. [2](#) 861
- [38] Arnur Nigmatov and Dmitriy Morozov. Topological optimization with big steps. *Discrete & Computational Geometry*, 2024. [3](#) 862
- [39] Jeong Joon Park, Peter Florence, Julian Straub, Richard Newcombe, and Steven Lovegrove. DeepSDF: Learning continuous signed distance functions for shape representation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 165–174, 2019. [2](#) 863
- [40] Songyou Peng, Chiyu Jiang, Yiyi Liao, Michael Niemeyer, Marc Pollefeys, and Andreas Geiger. Shape as points: A differentiable Poisson solver. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. [2](#) 864
- [41] Adrien Poulenard, Primož Skraba, and Maks Ovsjanikov. Topological function optimization for continuous shape matching. *Computer Graphics Forum*, 37(5):13–25, 2018. [2](#) 865
- [42] Andrei Sharf, Thomas Lewiner, Gil Shklarski, Sivan Toledo, and Daniel Cohen-Or. Interactive topology-aware surface reconstruction. *ACM Transactions on Graphics*, 26(3):43:1–43:10, 2007. [3](#) 866
- [43] Tianchang Shen, Jun Gao, Kangxue Yin, Ming-Yu Liu, and Sanja Fidler. Deep marching tetrahedra: a hybrid representation for high-resolution 3D shape synthesis. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. [2](#) 867
- [44] Tianchang Shen, Jacob Munkberg, Jon Hasselgren, Kangxue Yin, Zian Wang, Wenzheng Chen, Zan Gojcic, Sanja Fidler, Nicholas Sharp, and Jun Gao. Flexible isosurface extraction for gradient-based mesh optimization. *ACM Transactions on Graphics*, 42(4), 2023. [2](#) 868
- [45] Tianqi Shen, Shaohua Liu, Jiaqi Feng, Ziyi Ma, and Ning An. Topology-aware 3D Gaussian splatting: Leveraging persistent homology for optimized structural integrity. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 6823–6832, 2025. [3](#) 869
- [46] Kenji Tojo, Bernd Bickel, and Nobuyuki Umetani. DiffSoup: Direct differentiable rasterization of triangle soup for extreme radiance field simplification. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. arXiv:2603.27151. [1](#), [2](#), [3](#) 870
- [47] Dominik J. E. Waibel, Scott Atwell, Matthias Meier, Carsten Marr, and Bastian Rieck. Capturing shape information with multi-scale topological loss terms for 3D reconstruction. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, pages 150–159, 2022. [2](#) 871
- [48] Nanyang Wang, Yinda Zhang, Zhuwen Li, Yanwei Fu, Wei Liu, and Yu-Gang Jiang. Pixel2Mesh: Generating 3D mesh models from single RGB images. In *European Conference on Computer Vision (ECCV)*, 2018. [2](#) 872
- [49] Peng Wang, Lingjie Liu, Yuan Liu, Christian Theobalt, Taku Komura, and Wenping Wang. NeuS: Learning neural implicit surfaces by volume rendering for multi-view reconstruction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. [2](#) 873
- [50] Wang Yifan, Felice Serena, Shihao Wu, Cengiz Öztireli, and Olga Sorkine-Hornung. Differentiable 874

908 surface splatting for point-based geometry processing.
909 ACM Transactions on Graphics, 38(6), 2019. 2